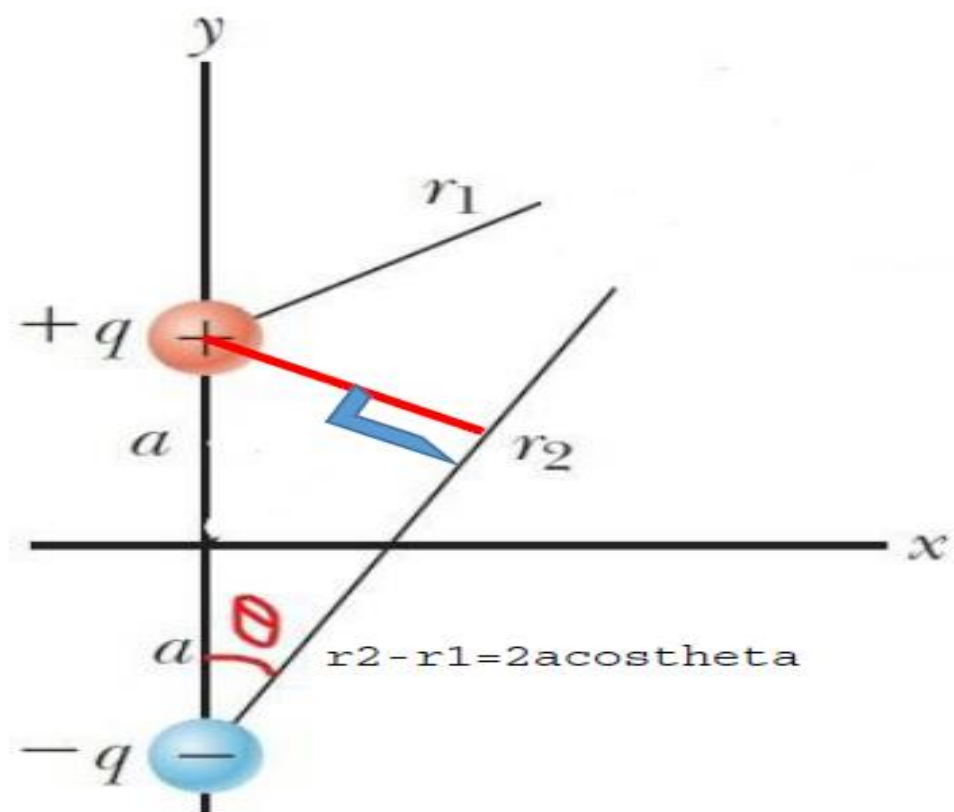
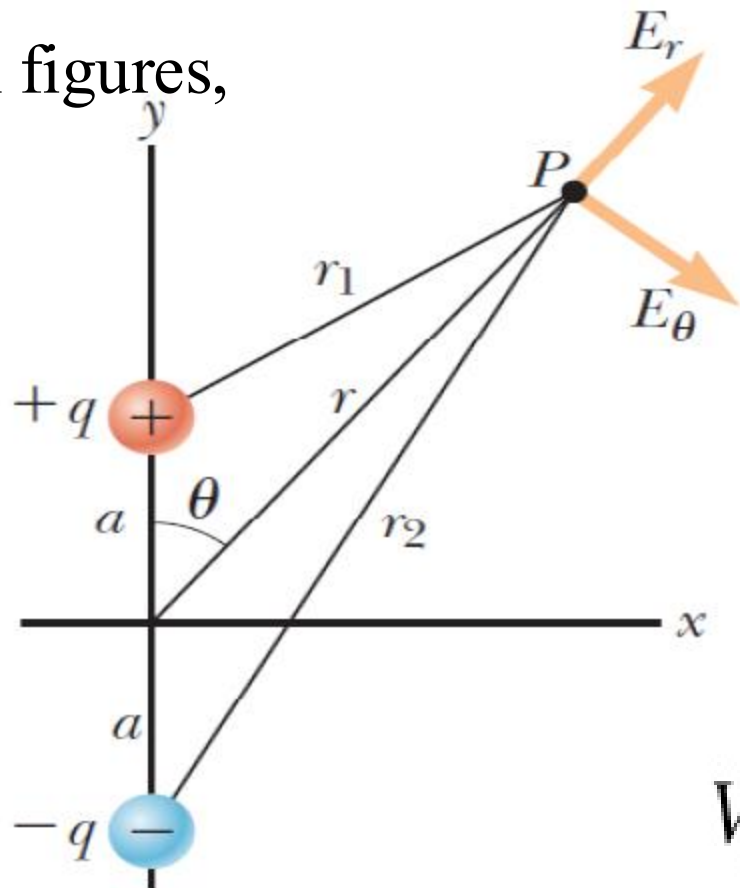


Electric potential due to Electric dipole

From figures,



$$V_P = V_{+q} + V_{-q} = k \left(\frac{q}{r_1} + \frac{-q}{r_2} \right) = kq \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

With $k = 1/4\pi\epsilon_0$

When $r \gg a$, then r_1 and r_2 are approximately equal and $r_2 - r_1 \approx 2a \cos \theta$, $r_1 r_2 \approx r^2$

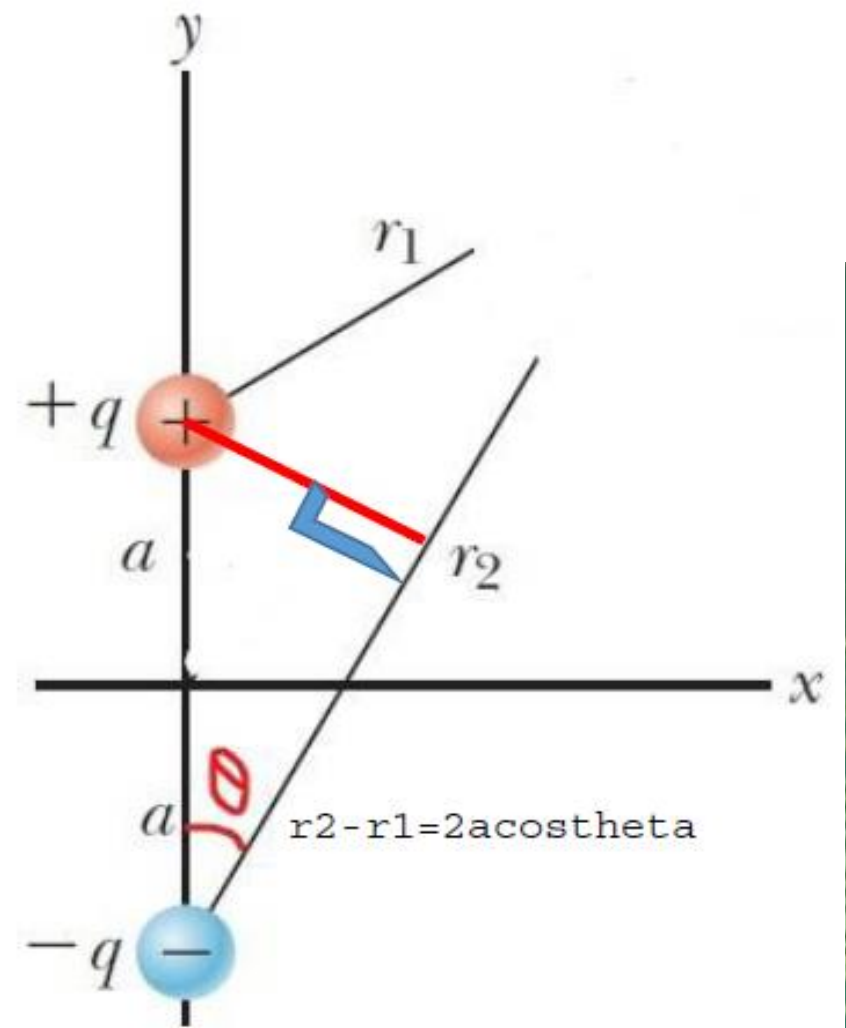
Therefore,
$$V_P = \frac{k2aq \cos \theta}{r^2} = k \frac{p \cos \theta}{r^2} = \left(k \frac{\vec{p} \cdot \vec{r}}{r^3} \right)$$

Note that the dipole potential decreases as $1/r^2$ whereas the electric field decreases as $1/r^3$.

Above equation gives the value of electric potential due to dipole at any point.

If $\theta = 0^\circ$, i.e. *At axial line*, $V = kp/r^2$

If $\theta = 90^\circ$, i.e. *At equatorial line* $V=0$, *That means it would require no work to bring a charge from infinity to the center of the dipole if it moves along the dipole bisector.*



Electric dipole in the Electric Field

An electric dipole with dipole moment $p = 2qa$ is placed in the uniform electric field E at an angle θ (angle between p and E). The force gain by the charges are shown in figure.

$$F_{\text{net}} = [qE + (-qE)] = 0 \text{ (Due to uniform Field).}$$

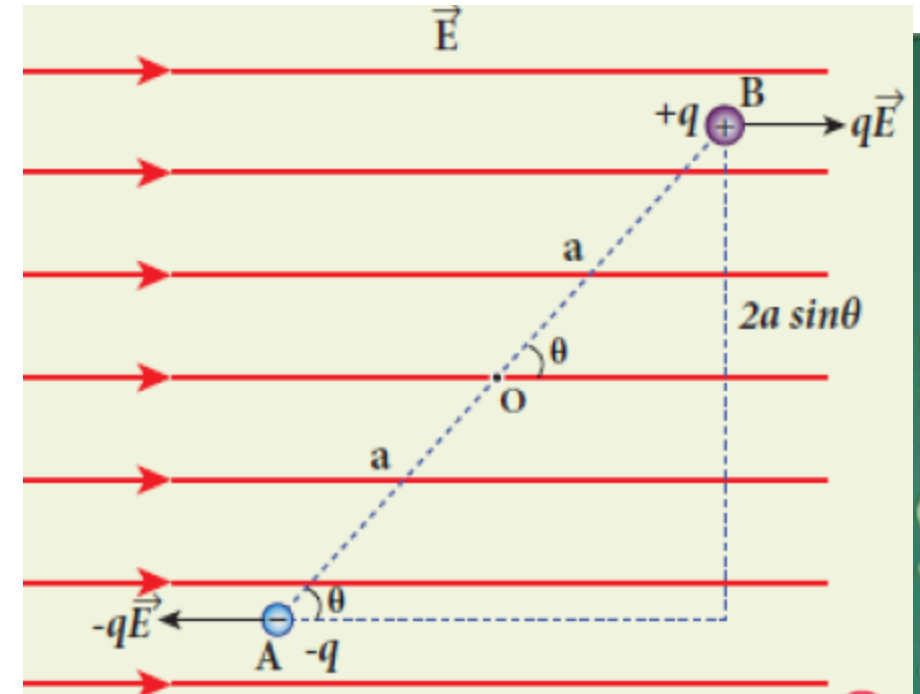
Two forces acting at different points will constitute a couple and the dipole experiences a torque, this torque tends to rotate the dipole along the direction of E field.

From figure, the magnitude of torque is

$$\tau = 2aqE \sin\theta. \quad (\text{along the perpendicular to the plane of the paper and directed into it}).$$

In Vector notation, $\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$ (where, $p = 2qa$)

$\tau = \tau_{\text{max}}$ when $\theta = 90^\circ$. and $\tau = 0$ when $\theta = 0^\circ$ (p is parallel with E)



Work done by the torque during small rotation $d\theta$ *against the electric field* is,

$$dw = - \int \tau d\theta$$

$$dw = - \int pE \sin\theta d\theta$$

But Change in potential energy of the system is , $dU = - dw$

$$dU = \int pE \sin\theta d\theta$$

Change in PE when dipole rotated from angle θ_1 to θ_2 is

$$U(\theta_2) - U(\theta_1) = \int_{\theta_1}^{\theta_2} pE \sin\theta d\theta$$

$$U(\theta_2) - U(\theta_1) = - pE(\cos\theta_2 - \cos\theta_1)$$

If the angle θ is changed from ($\theta_1=90^0$) to ($\theta_2= \theta$)

Then ,
$$U(\theta) - U(90^0) = -pE \cos \theta$$

If we choose Potential energy of the dipole to be Zero when $\theta= 90^0$
(Dipole axis is perpendicular to the field), $U(90^0)= 0$. then above equation becomes,

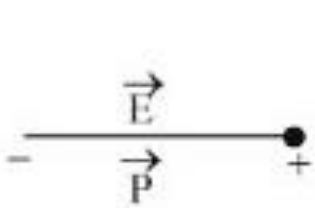
$$U(\theta) = -pE \cos \theta.$$

For example.

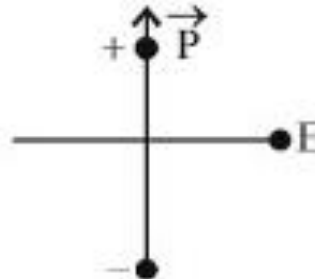
$$W_{\theta \rightarrow 180^\circ} = PE (1 - (-1)) = 2PE$$

$$W_{\theta \rightarrow 90^\circ} = PE (1 - 0) = PE$$

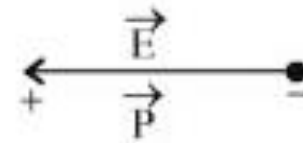
If a dipole is rotated from field direction ($\theta = 0$) to θ , then $W = PE (1 - \cos \theta)$



$$\tau = \min = 0$$
$$W = \min = 0$$



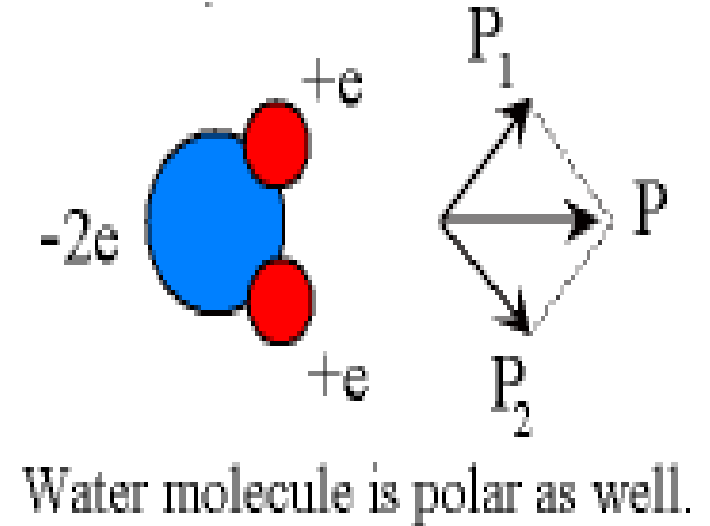
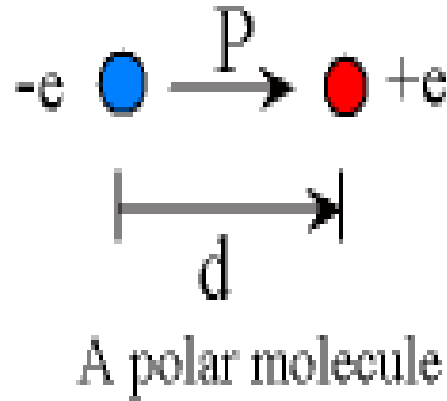
$$\tau = \max. = PE$$
$$W = PE$$



$$\tau = \min = 0$$
$$W = \max. = 2PE$$

Application

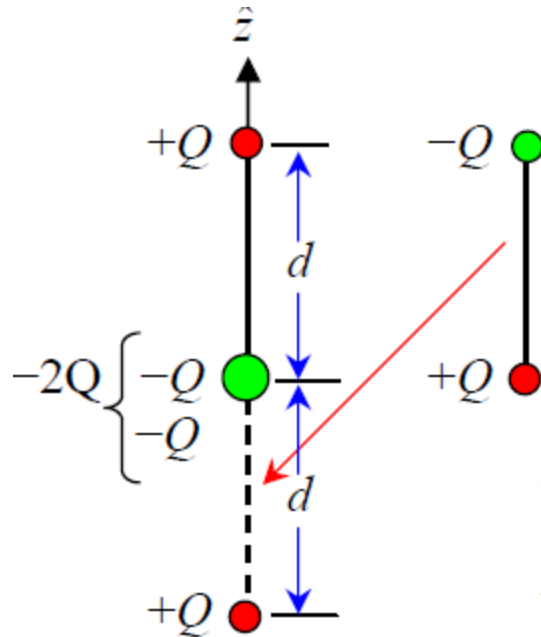
Microwave oven



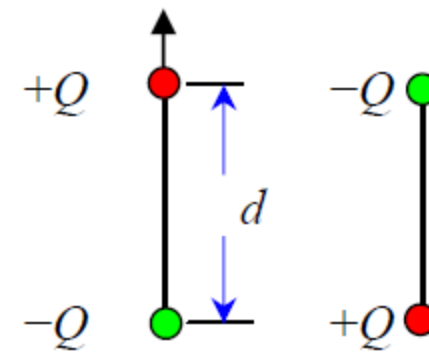
Microwave oven works on the principle of torque acting on an electric dipole. The food we consume has water molecules which are permanent electric dipoles. Oven produces microwaves that are oscillating electromagnetic fields and produce torque on the water molecules. Due to this torque on each water molecule, the molecules rotate very fast and produce thermal energy. Thus, heat generated is used to heat the food.

Electric Quadrupole

The arrangement of *four equal and opposite charges* or two dipoles arrangement (*dipole moments are in opposite directions*) with certain electric field at any point is called a electric quadrupole. There are different types of quadrupole as shown below. The Quadrupole moment from the figure is given by $Q_0 = 2Qd^2$. It is **Tensor quantity**.



Linear Quadrupole



Square Quadrupole

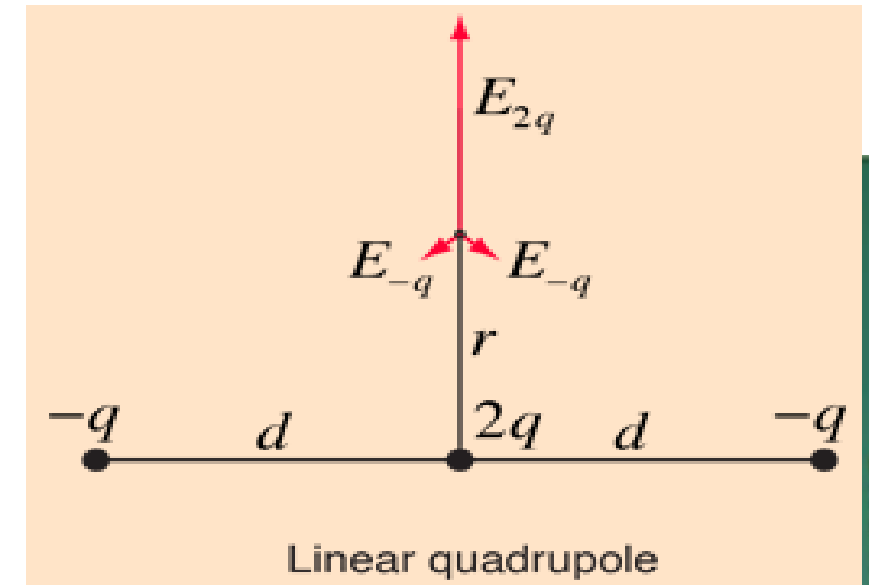
Electric field due to linear Quadrupole *At Equatorial point:*

$$E_{2q} = \frac{2q}{4\pi\epsilon_0 r^2}$$

$$E_{-q} = \frac{-q}{4\pi\epsilon_0 (r^2 + d^2)}$$

By symmetry, the field components parallel to the quadrupole cancel.
The resultant field perpendicular is

$$E_r = \frac{2q}{4\pi\epsilon_0} \left[\frac{1}{r^2} - \frac{r}{(r^2 + d^2)^{3/2}} \right]$$



Factoring out the $\frac{1}{r^2}$ term puts it in the form

$$E_r = \frac{2q}{4\pi\epsilon_0 r^2} \left[1 - \left(1 + \frac{d^2}{r^2} \right)^{-3/2} \right]$$

For $\frac{d^2}{r^2} \ll 1$, the binomial expansion gives

$$\left(1 + \frac{d^2}{r^2} \right)^{-3/2} \approx 1 - \frac{3}{2} \frac{d^2}{r^2}$$

and the quadrupole electric field at large distances simplifies to

$$E_r \approx \frac{3qd^2}{4\pi\epsilon_0 r^4}$$

Electric Field due to Line Quadrupole

Along Axial Line ???

HOMEWORK

For $r \gg l$, using Binomial theorem

$$(1 + x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \dots$$

Here $x = \left(\frac{l^2}{r^2} + \frac{2l}{r} \cos \theta\right) \ll 1$ and $n = -\frac{1}{2}$, On Solving Equation (3) Becomes,

$$\frac{1}{|r + l|} = \frac{1}{r} \left[1 + \frac{l^2}{r^2} + \frac{2l}{r} \cos \theta \right]^{-1/2}$$

$$\frac{1}{|r + l|} = \frac{1}{r} \left[1 - \frac{l^2}{2r^2} - \frac{l}{r} \cos \theta + \frac{3}{8} \left(\frac{l^2}{r^2} + \frac{2l}{r} \cos \theta \right)^2 \right]$$

$$\frac{1}{|r + l|} = \frac{1}{r} \left[1 + \frac{l^2}{2r^2} (3\cos^2 \theta - 1) - \frac{l}{r} \cos \theta \right]$$

$$\frac{1}{|r - l|} = \frac{1}{r} \left[1 + \frac{l^2}{2r^2} (3\cos^2 \theta - 1) + \frac{l}{r} \cos \theta \right]$$

Solving with r^3 and Higher order $\rightarrow 0$

Then from Equation (1)

$$V = \frac{q}{4\pi\epsilon_0 r} \left[1 + \frac{l^2}{2r^2} (3\cos^2\theta - 1) + \frac{l}{r} \cos\theta \right] - 2 + \left[1 + \frac{l^2}{2r^2} (3\cos^2\theta - 1) - \frac{l}{r} \cos\theta \right]$$

$$V = \frac{ql^2}{4\pi\epsilon_0 r^3} (3\cos^2\theta - 1)$$

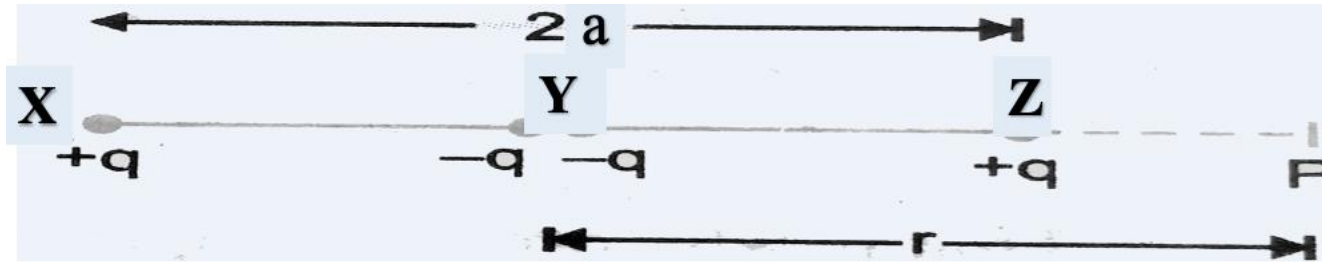
At $\theta=0$ (I.e. At Axial Line), Potential due to Quadrupole is

$$V = \frac{2ql^2}{4\pi\epsilon_0 r^3} = \frac{Q_0}{4\pi\epsilon_0 r^3} \quad (\text{with } Q_0 = 2ql^2 \text{ Quadrupole Moment})$$

At $\theta = 90$ (I.e. At Equatorial Line), Potential due to Quadrupole is

$$V = -\frac{ql^2}{4\pi\epsilon_0 r^3}$$

Electric potential due to Quadrupole along axial line (alternative method)



A point is located at P, Which is r distance away from point B. The system of charges forms an electric quadrupole. It can be considered that the system of the electric quadrupole has three charges.

Charge $+q$ placed at point A, Charge $-2q$ placed at point B, Charge $+q$ placed at point C.

$AB=BC=d$, $BP=r$

Electrostatic potential caused by the system of three charges at point P is given by,

$$\begin{aligned}
 V &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{XP} - \frac{2q}{YP} + \frac{q}{ZP} \right] \\
 &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r+a} - \frac{2q}{r} + \frac{q}{r-a} \right] \\
 &= \frac{Q}{4\pi\epsilon_0} \left[\frac{r(r-a) - 2(r+a)(r-a) + r(r+a)}{r(r+a)(r-a)} \right] \\
 &= \frac{q}{4\pi\epsilon_0} \left[\frac{r^2 - ra - 2r^2 + 2a^2 + r^2 + ra}{r(r^2 - a^2)} \right] = \frac{q}{4\pi\epsilon_0} \left[\frac{2a^2}{r(r^2 - a^2)} \right] \\
 &= \frac{2qa^2}{4\pi\epsilon_0 r^3 \left(1 - \frac{a^2}{r^2} \right)}
 \end{aligned}$$

Since $\frac{r}{a} \gg 1$

$\therefore \frac{a}{r} \ll 1$

$\frac{a^2}{r^2}$ is taken as negligible.

$$\therefore V = \frac{2qa^2}{4\pi\epsilon_0 r^3}$$

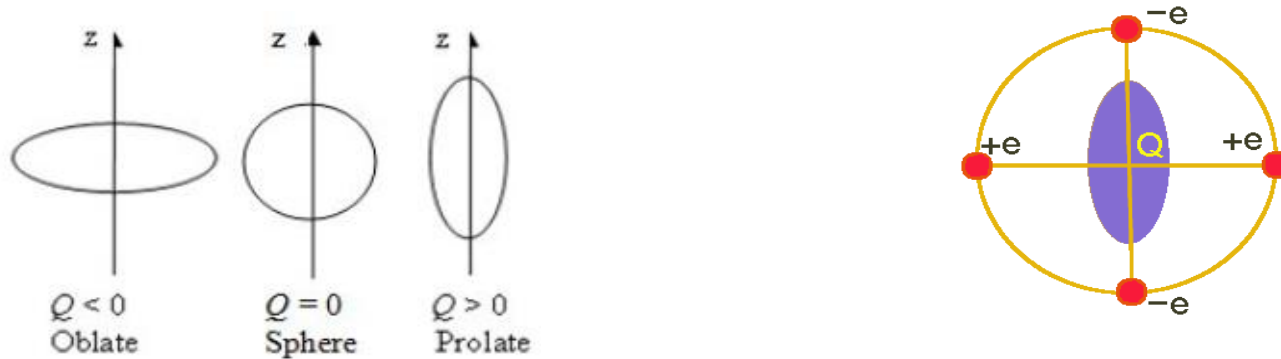
It can be inferred that potential, $V \propto \frac{1}{r^3}$

However, it is known that for a diopole, $V \propto \frac{1}{r^3}$

And, for a monopole, $V \propto \frac{1}{r}$

Significance of Quadrupole moment.

The *nuclear quadrupole moment* is a parameter which describes the effective *shape of the ellipsoid of nuclear charge distribution*. A non-zero quadrupole moment Q indicates that the charge distribution is not spherically symmetric. By convention, the value of Q is taken to be positive if the ellipsoid is prolate and negative if it is oblate as shown in figure.



In other words 'Q' represents the departure from spherical symmetry in the rest frame of the nucleus. If a nucleus is not spherically symmetric, it will have a non-zero electric quadrupole moment, so the measurement of the quadrupole moment is a test of the shell theory.

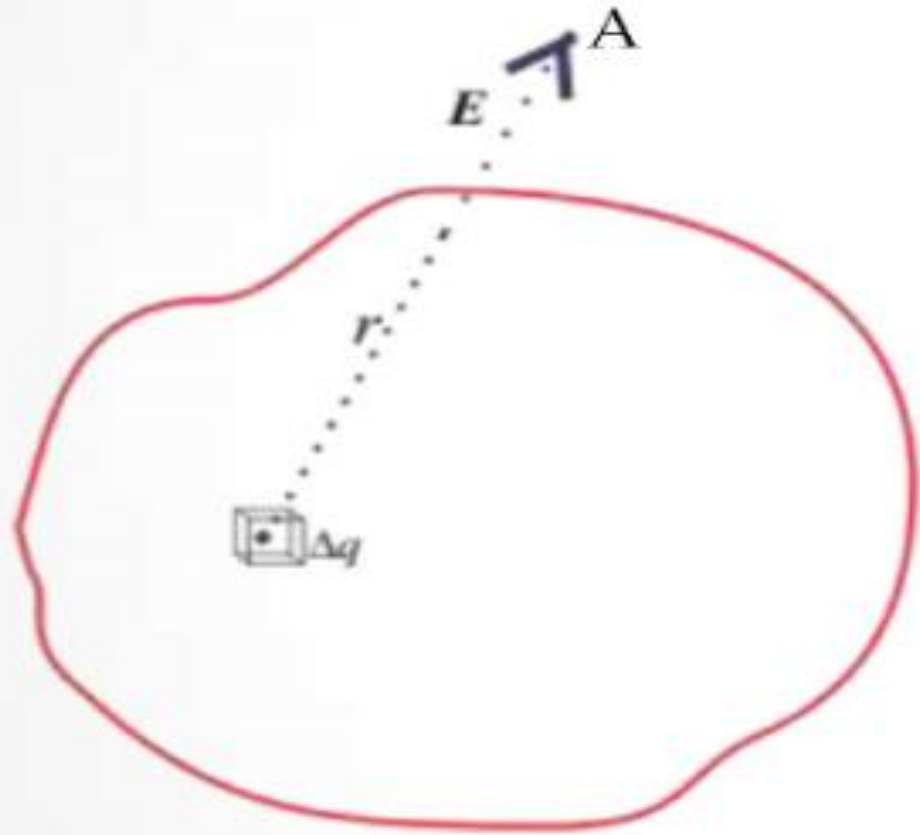
The quadrupole moment is essential when the dipole moment vanishes but the system still has **nontrivial electric or magnetic effects** due to spatial charge configuration.

Many molecules like **CO₂**, **benzene**, and **O₂** have **no dipole moment** but do have **quadrupole moments**.

Electric field due to continuous charge distributions

- Charges on surface of system are located very close together and can be assumed to have continuous distribution of charges.
- In a system of closely spaced charges charge could be continuously distributed along
 - ❑ Some line (Length)
 - ❑ Over a surface (Area)
 - ❑ Throughout a Volume (Volume)

Charged system with continuous distribution of charge can be divided into very small elements with charge Δq as shown below in the figure



Electric field at A due to charged element Δq is

$$\Delta E = \frac{1}{4\pi\epsilon_0} \frac{\Delta q}{r^2} \hat{r}$$

Total field at point A due to all the charge elements

$$E \cong \frac{1}{4\pi\epsilon_0} \sum_i \frac{\Delta q_i}{(r_i)^2} \hat{r}_i$$

For charge distributed continuously over some region, the sum become integral . So total field at point A within limit $\Delta q \rightarrow 0$ is

$$E = \frac{1}{4\pi\epsilon_0} \lim_{\Delta q \rightarrow 0} \sum_i \frac{\Delta q_i}{(r_i)^2} \hat{r}_i = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$

This integral is done over the entire charge distribution.

Line Charge

When the distribution of charge is uniformly along the line then it is called Linear Charge Distribution.

Line Charge Density λ ,

$$\lambda = \frac{Q}{L}$$

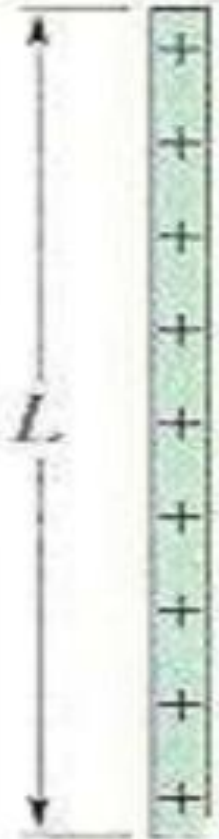
Q = charge uniformly distributed over line

L = length of the line

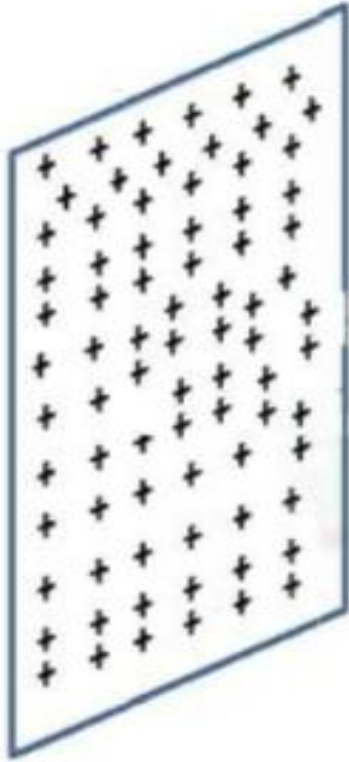
Unit of λ is Coulomb/meter (C/m)

For non-uniform distribution of charge

$$\lambda = \frac{dQ}{dL}$$



Surface Charge



When the charge distribution is over a particular area then the distribution is called as Surface charge distribution. S Surface Charge Density σ ,

$$\sigma = \frac{Q}{A}$$

Q = charge uniformly distributed over surface of area A

Unit of σ is C/m^2

For non-uniform charge distributions

$$\sigma = \frac{dQ}{dA}$$

Volume Charge

When the charge distribution is over a particular volume then the distribution is called as Volume charge distribution. Volume Charge Density ρ ,

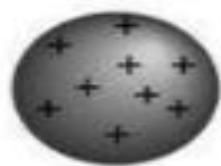
$$\rho = \frac{Q}{V}$$

Q = charge uniformly distributed over system of volume V

Unit of ρ is C/m^3

For non-uniform charge distributions

$$\rho = \frac{dQ}{dV}$$



Electric field due to Charged Ring

Small charge due small length element ds is, $dq = \lambda ds$,
Electric Field due to this charge at point P is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2}.$$

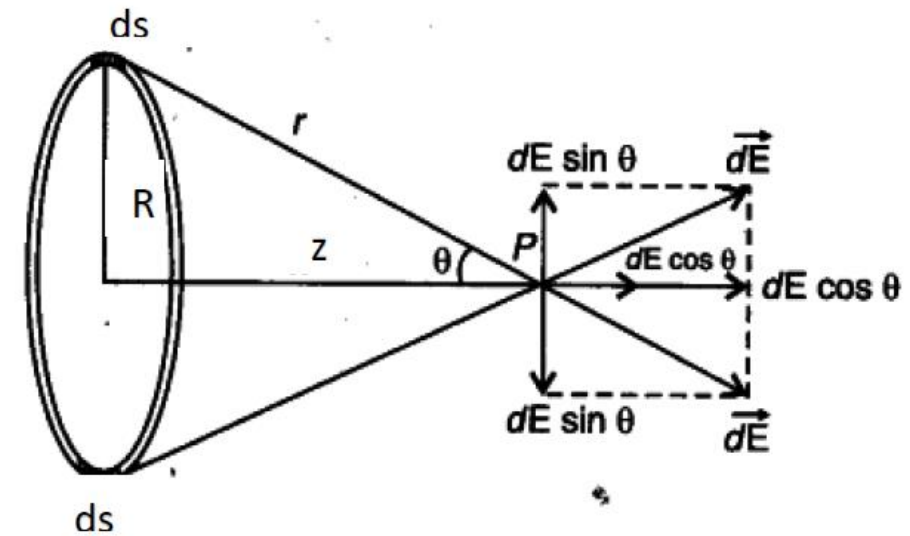
$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{(z^2 + R^2)}.$$

$$\cos \theta = \frac{z}{r} = \frac{z}{(z^2 + R^2)^{1/2}}.$$

$$dE \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{z\lambda}{(z^2 + R^2)^{3/2}} ds.$$

$$E = \int dE \cos \theta = \frac{z\lambda}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} \int_0^{2\pi R} ds$$
$$= \frac{z\lambda(2\pi R)}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}.$$

$$E = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}$$



From Figure, Vertical components ($\sin \theta$) cancel each other and horizontal component ($\cos \theta$) contributes to the net electric field.

If $z = 0$ (i.e. at center of the ring) $E=0$.that means at the center of the ring there is no electric field.

If, $z \gg R$. For such a point, the expression $z^2 + R^2$ can be approximated as z^2 , and becomes

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2}$$

The magnitude of Electric field becomes maximum when, $\frac{dE}{dz} = 0$. then

$$\frac{dE}{dz} = \frac{q}{4\pi\epsilon_0} \frac{d}{dz} \left[\frac{z}{(R^2 + Z^2)^{\frac{3}{2}}} \right]$$

$$0 = \frac{1}{(R^2 + Z^2)^{3/2}} + Z \left\{ \left(-\frac{3}{2}\right) (R^2 + Z^2)^{-\frac{5}{2}} \times 2Z \right\}$$

$$0 = \frac{1}{(R^2 + Z^2)^{\frac{3}{2}}} - 3Z^2 (R^2 + Z^2)^{-5/2}$$

$$3Z^2 = R^2 + Z^2$$

$$Z = \pm \frac{R}{\sqrt{2}}$$

Maximum value of Electric field becomes,

$$|E| = E_0 = \frac{qZ}{4\pi\epsilon_0 (Z^2 + R^2)^{3/2}} = \frac{q \frac{R}{\sqrt{2}}}{4\pi\epsilon_0 \left(\frac{R^2}{2} + R^2\right)^{3/2}} = \frac{qR}{4\pi\epsilon_0 R^3 \sqrt{2} \left(\frac{1}{2} + 1\right)^{\frac{3}{2}}} = \frac{qR}{4\pi\epsilon_0 R^3 \sqrt{2} \left(\frac{3\sqrt{3}}{2\sqrt{2}}\right)}$$

A plot is shown below.

$$E_0 = \frac{q}{6\sqrt{3}\pi\epsilon_0 R^2}$$

If $R \gg Z$ then

From Equation

$$E = \frac{1}{4\pi\epsilon_0} \left[\frac{qz}{(R^2 + Z^2)^{\frac{3}{2}}} \right]$$

$$E = \left[\frac{q}{4\pi\epsilon_0 R^3} \right] Z$$

If a $(-Q)$ charge is placed near the center Of this charged Ring (**NOT AT EXACTLY CENTER, AT CENTER $E=0$**)

This charge gets Electric force as $F = -QE$
Charge undergoes the SHM as shown in Graph. Comparing with equation of SHM then frequency of oscillation is given by

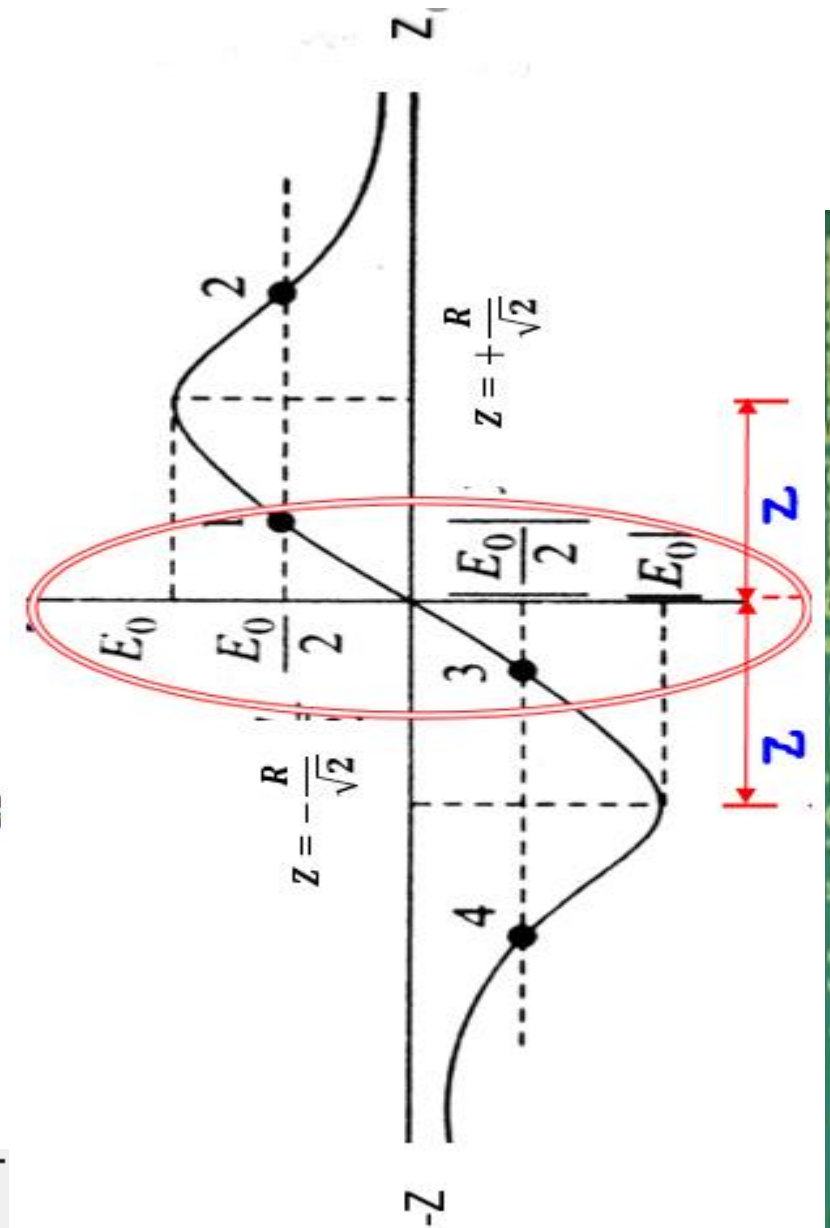
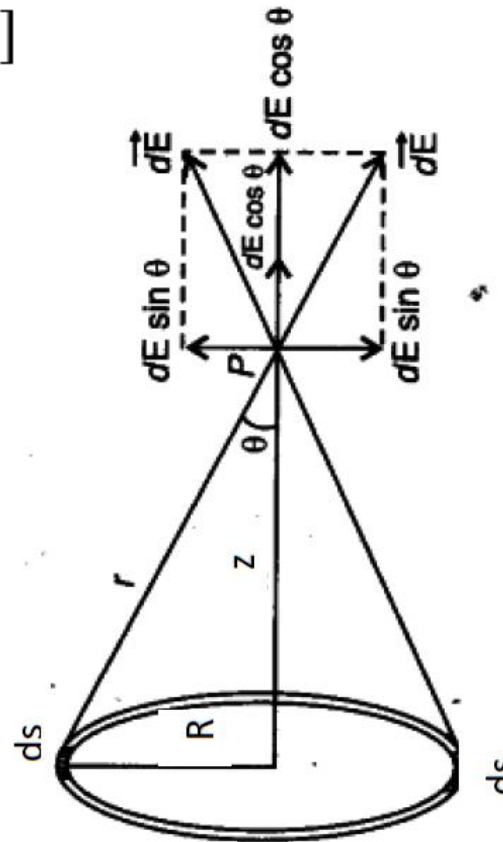
$$F = \left[\frac{-Qq}{4\pi\epsilon_0 R^3} \right] Z$$

$$M_Q \frac{d^2 Z}{dt^2} = \left[\frac{-Qq}{4\pi\epsilon_0 R^3} \right] Z$$

Comparing with equation of SHM then frequency of oscillation is given by

Where $M_Q =$ mass of charge

$$\omega = \sqrt{\left[\frac{Qq}{4\pi M_Q \epsilon_0 R^3} \right]}$$



Electric Potential due to Charged Ring

Small differential length element $dl=Rd\phi'$

Charge this element is $dq=\lambda dl=\lambda R d\phi'$

Potential due to this element at point P is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda R d\phi'}{\sqrt{R^2 + z^2}}$$

Total potential due to entire ring is

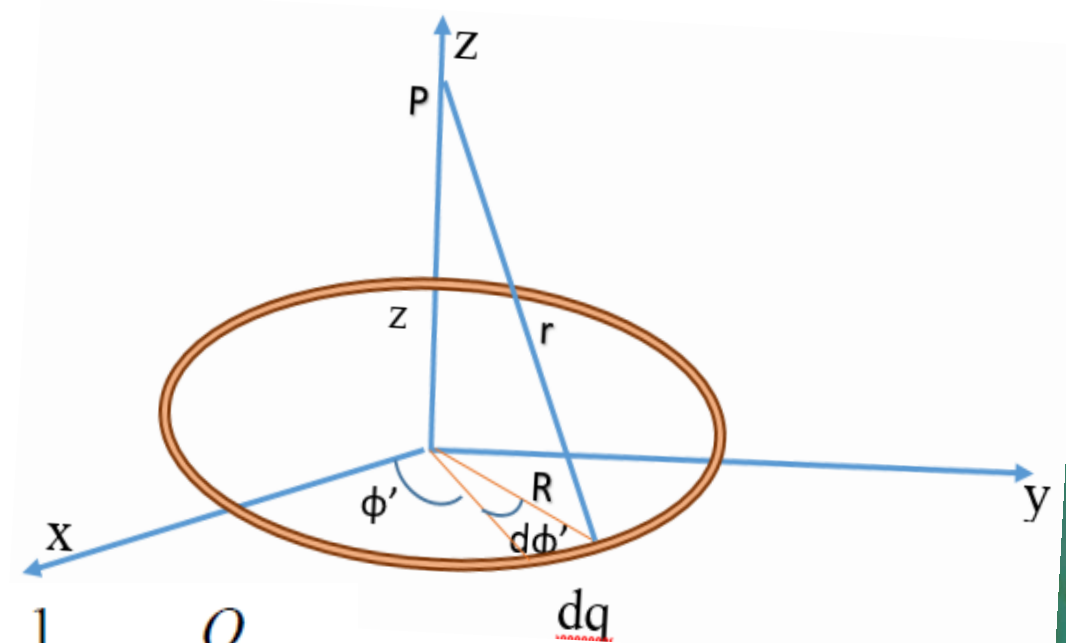
$$V = \int dV = \frac{1}{4\pi\epsilon_0} \frac{\lambda R}{\sqrt{R^2 + z^2}} \oint_0^{2\pi} d\phi' = \frac{1}{4\pi\epsilon_0} \frac{2\pi\lambda R}{\sqrt{R^2 + z^2}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{R^2 + z^2}}$$

where we have substituted $Q=2R\pi\lambda$ for the total charge on the ring.

In the limit $Z \gg R$, the potential approaches ***“point-charge” limit as***

Corresponding electric field obtained as,

$$E_z = -\frac{\partial V}{\partial z} = -\frac{\partial}{\partial z} \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{R^2 + z^2}} \right) = \frac{1}{4\pi\epsilon_0} \frac{Qz}{(R^2 + z^2)^{3/2}}$$



$$V \approx \frac{1}{4\pi\epsilon_0} \frac{Q}{z}$$

Electric field due to Charged Disk

From figure R = Radius of Disk.

$\sigma = q/\pi R^2$ = Surface charge density of the disk.

We want to calculate the E field at point P at Distance z from the center of the disk. We consider The disk as a set of concentric rings. radius and dr thickness of such one ring.

Surface area of the ring $= 2\pi r dr$

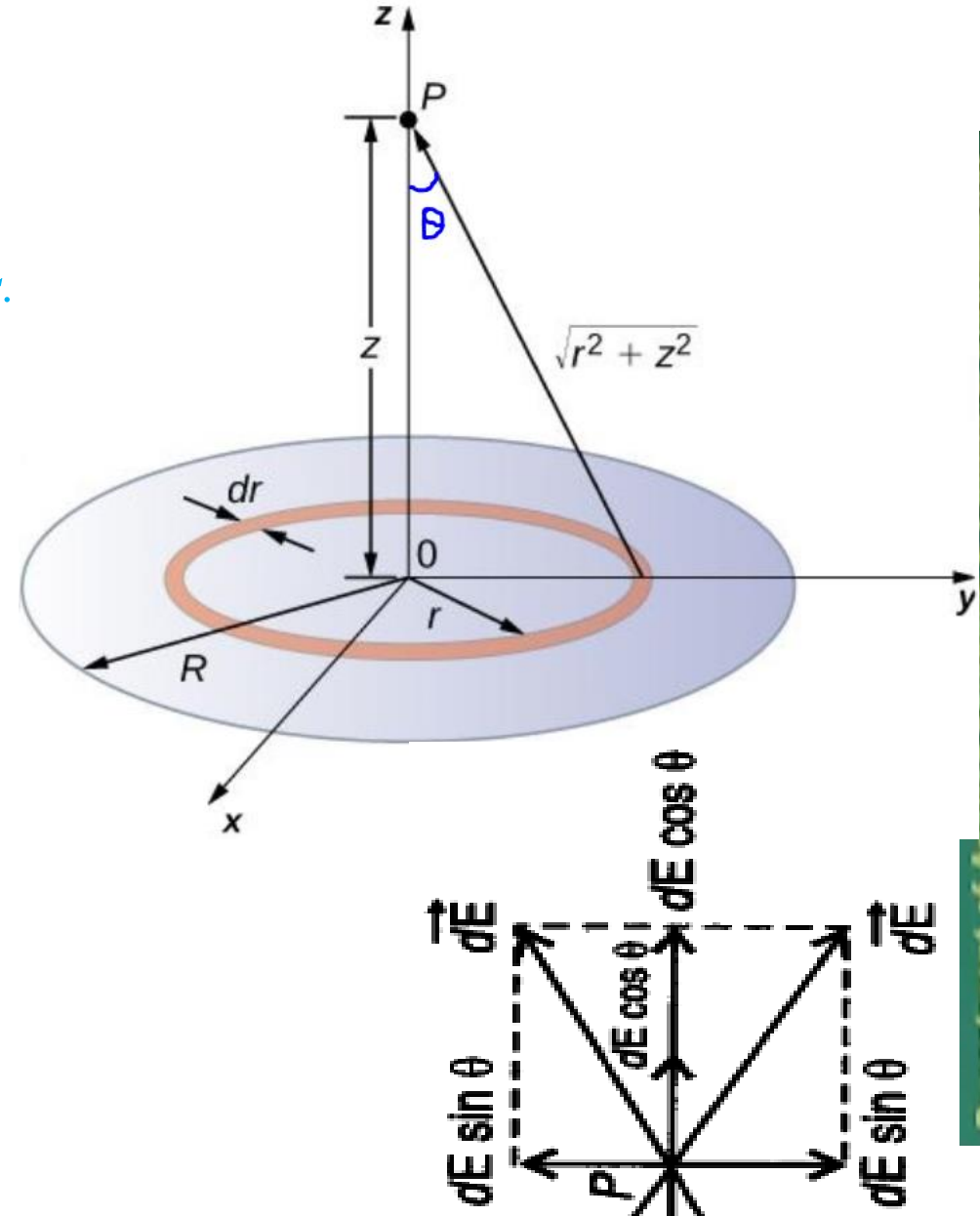
Charge on the ring (dq) $= \sigma 2\pi r dr$

Electric field at Point P due to this charge is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi r dr}{(r^2 + z^2)^{3/2}}$$

This field has two components Horizontal and Vertical as shown in figure. Horizontal components cancel Each other only vertical component $dE \cos \theta$ contributes to electric field.

Total electric field is given by $E = \int_0^R dE \cos \theta$



$$E = \int dE \cos\theta = \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi r dr}{(r^2 + z^2)} \frac{z}{\sqrt{r^2 + z^2}} \Rightarrow E = \frac{\pi\sigma z}{4\pi\epsilon_0} \int_0^R \frac{2r dr}{(z^2 + r^2)^{3/2}}$$

$$E = \frac{\sigma z}{4\epsilon_0} \int_0^R (z^2 + r^2)^{-\frac{3}{2}} (2r) dr, \quad \text{using integral } \int X^m dX = \frac{X^{m+1}}{m+1},$$

With $X=(z^2+r^2)$, $m = -3/2$, and $dX=(2r)dr$

We get, $E = \frac{\sigma z}{4\epsilon_0} \left[\frac{(z^2 + r^2)^{-1/2}}{-\frac{1}{2}} \right]_0^R$. After rearranging the terms ,

If $R \rightarrow \infty$ (Infinite charged sheet) and z is finite then,

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

$$E = \frac{\sigma}{2\epsilon_0}$$

Electric potential due to Charged Disk

Surface area of the ring $= 2\pi r dr$

Charge on 1

$$V \approx \frac{\sigma}{2\epsilon_0} \cdot \frac{R^2}{2|z|} = \frac{1}{4\pi\epsilon_0} \frac{\sigma(\pi R^2)}{|z|} = \frac{1}{4\pi\epsilon_0} \frac{Q}{|z|}$$

Electric field at Point P due to this charge is , $dV = \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi r dr}{(r^2 + z^2)^{1/2}}$

Total electric potential is, $V = \int_0^R dV = \frac{\sigma}{2\epsilon_0} \int_0^R \frac{r dr}{\sqrt{r^2 + z^2}}$

$$V = \frac{\sigma}{2\epsilon} [\sqrt{r^2 + z^2}]_0^R \implies V = \frac{\sigma}{2\epsilon} (\sqrt{R^2 + z^2} - z).$$

In the limit $z \gg R$, then,

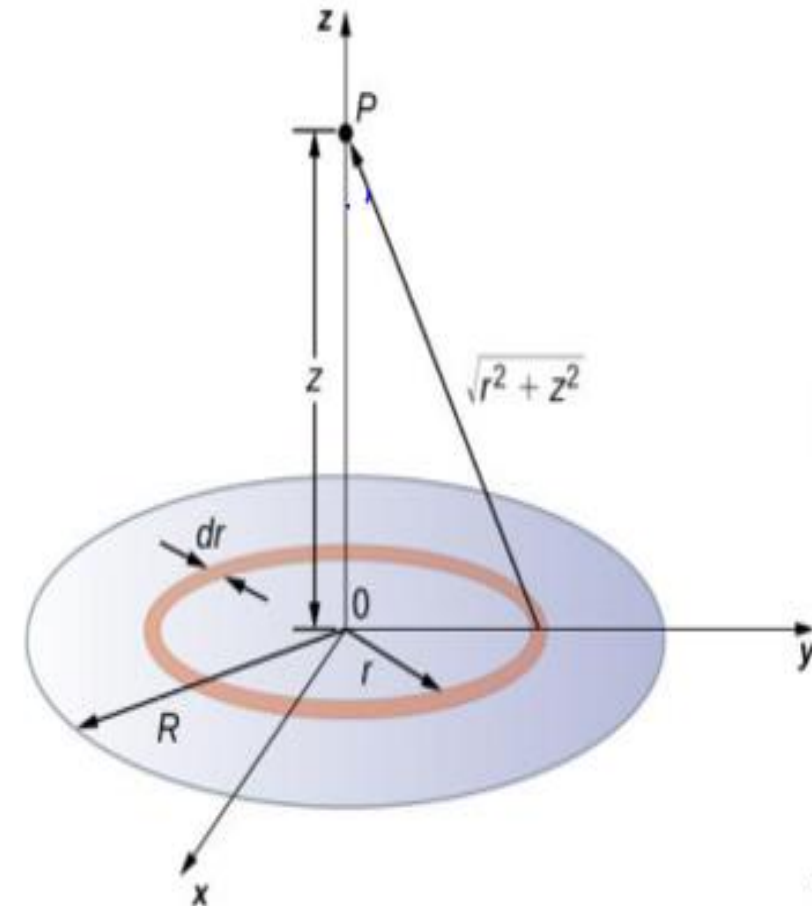
$$\sqrt{R^2 + z^2} = |z| \left(1 + \frac{R^2}{z^2} \right)^{1/2} = |z| \left(1 + \frac{R^2}{2z^2} + \dots \right)$$

and the potential simplifies to the point-charge limit

Corresponding electric field is,

$$E_z = -\frac{\partial V}{\partial z} = -\frac{\sigma}{2\epsilon_0} \frac{d}{dz} (\sqrt{z^2 + R^2} - z)$$

$$= \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right).$$



See the Summary of the field and potential

THANK YOU



summary of field and potential.jpg